

Course Practical Project (100 Marks

Group members

- | | |
|-----------------------------|-----------|
| ➤ Thando Mtsweni | 222225246 |
| ➤ Happyness Mazibuko | 222031778 |
| ➤ Mkhwanazi cashile Kwanele | 220465738 |
| ➤ Nozethu Maseko | 201991322 |

Overview

In this project, a mini-application named the *Illegal Dumping Complaint Tracker* is developed using MongoDB as the main NoSQL database. Users can report and handle instances of illegal dumping inside a municipality according to the system's architecture.

This project aims to show how NoSQL database principles like arrays, aggregation pipelines, and embedded documents are used. Additionally, the system supports complete CRUD activities and offers a graphical user interface.

System Overview

The web-based interface of the system enables users to:

- Report new concerns
- View current complaints
- Add status updates to complaints.
- Include references to images
- Generate reports using data that has been stored.

The program is constructed with:

MongoDB for storing data For backend services, use Node.js and Express.

For the front-end interface, use React (Vite).

Conclusion

The Illegal Dumping Complaint Tracker shows how MongoDB may be used practically to create a real-world solution. Complex data may be stored with flexibility thanks to the usage of NoSQL structures, and the aggregation pipeline offers strong data analysis capabilities.

The system demonstrates the usefulness of MongoDB in contemporary application development and satisfies every criteria of the Advanced Database practical project.